Java Technical Interview Ques and ans

1. Reverse a String

```
public class ReverseString {
  public static void main(String[] args) {
    String str = "Automation";
    StringBuilder reversed = new StringBuilder(str).reverse();
    System.out.println(reversed);
  }
}
```

2. Check for Palindrome

```
public class Palindrome {
  public static void main(String[] args) {
    String str = "madam";
    String reversed = new StringBuilder(str).reverse().toString();
    System.out.println(str.equals(reversed));
  }
}
```

3. Fibonacci Series

```
public class Fibonacci {
  public static void main(String[] args) {
    int n = 10, num1 = 0, num2 = 1;
    System.out.print("Fibonacci Series: " + num1 + ", " + num2);
    for (int i = 2; i < n; i++) {
      int num3 = num1 + num2;
      System.out.print(", " + num3);
      num1 = num2;
      num2 = num3;
    }
  }
```

```
}
```

4. Factorial of a Number

```
public class Factorial {
  public static void main(String[] args) {
    int num = 5, factorial = 1;
    for (int i = 1; i <= num; i++) {
      factorial *= i;
    }
    System.out.println(factorial);
  }
}
```

5. Prime Number Check

```
public class PrimeCheck {
  public static void main(String[] args) {
    int num = 11;
    boolean isPrime = true;
    for (int i = 2; i <= Math.sqrt(num); i++) {
      if (num % i == 0) {
        isPrime = false;
        break;
      }
    }
    System.out.println(isPrime);
  }
}
```

6. Count Vowels and Consonants

```
public class VowelConsonantCount {
  public static void main(String[] args) {
    String str = "Automation";
```

```
    int vowels = 0, consonants = 0;
    for (char c : str.toCharArray()) {
      if ("aeiouAEIOU".indexOf(c) != -1) {
        vowels++;
      } else if (Character.isLetter(c)) {
        consonants++;
      }
    }
    System.out.println("Vowels: " + vowels + ", Consonants: " + consonants);
  }
}
```

7. Sort an Array

```
import java.util.Arrays;
public class SortArray {
  public static void main(String[] args) {
    int[] arr = {5, 2, 8, 1, 3};
    Arrays.sort(arr);
    System.out.println(Arrays.toString(arr));
  }
}
```

8. Merge Two Arrays

```
import java.util.Arrays;
public class MergeArrays {
  public static void main(String[] args) {
    int[] arr1 = {1, 3, 5};
    int[] arr2 = {2, 4, 6};
    int[] merged = new int[arr1.length + arr2.length];
    System.arraycopy(arr1, 0, merged, 0, arr1.length);
    System.arraycopy(arr2, 0, merged, arr1.length, arr2.length);
```

```
    System.out.println(Arrays.toString(merged));
  }
}
```

9. Find the Largest Element in an Array

```
public class LargestInArray {
  public static void main(String[] args) {
    int[] arr = {1, 3, 5, 7, 9};
    int largest = arr[0];
    for (int num : arr) {
      if (num > largest) {
        largest = num;
      }
    }
    System.out.println(largest);
  }
}
```

10. Remove Duplicates from an Array

```
import java.util.HashSet;
public class RemoveDuplicates {
  public static void main(String[] args) {
    int[] arr = {1, 2, 2, 3, 4, 4};
    HashSet<Integer> set = new HashSet<>();
    for (int num : arr) {
      set.add(num);
    }
    System.out.println(set);
  }
}
```

11. Check if a Number is Armstrong

```
public class ArmstrongNumber {
  public static void main(String[] args) {
    int num = 153, sum = 0, temp = num;
    while (temp != 0) {
      int digit = temp % 10;
      sum += Math.pow(digit, 3);
      temp /= 10;
    }
    System.out.println(num == sum);
  }
}
```

12. Reverse a Number

```
public class ReverseNumber {
  public static void main(String[] args) {
    int num = 12345, reversed = 0;
    while (num != 0) {
      reversed = reversed * 10 + num % 10;
      num /= 10;
    }
    System.out.println(reversed);
  }
}
```

13. Calculate GCD of Two Numbers

```
public class GCD {
  public static void main(String[] args) {
    int a = 60, b = 48;
    while (b != 0) {
      int temp = b;
```

```
        b = a % b;
        a = temp;
      }
      System.out.println(a);
    }
}
```

14. Check for Anagram

```
import java.util.Arrays;
public class AnagramCheck {
  public static void main(String[] args) {
    String str1 = "listen", str2 = "silent";
    char[] arr1 = str1.toCharArray();
    char[] arr2 = str2.toCharArray();
    Arrays.sort(arr1);
    Arrays.sort(arr2);
    System.out.println(Arrays.equals(arr1, arr2));
  }
}
```

15. Count the Number of Digits in a Number

```
public class CountDigits {
  public static void main(String[] args) {
    int num = 12345;
    int count = String.valueOf(num).length();
    System.out.println(count);
  }
}
```

16. Print the Prime Numbers in a Range

```
public class PrimeInRange {
  public static void main(String[] args) {
```

```
    int start = 10, end = 50;
    for (int num = start; num <= end; num++) {
      boolean isPrime = true;
      for (int i = 2; i <= Math.sqrt(num); i++) {
        if (num % i == 0) {
          isPrime = false;
          break;
        }
      }
      if (isPrime && num > 1) {
        System.out.print(num + " ");
      }
    }
  }
}
```

17. Find the Second Largest Element in an Array

```
public class SecondLargest {
  public static void main(String[] args) {
    int[] arr = {12, 35, 1, 10, 34, 1};
    int first = Integer.MIN_VALUE, second = Integer.MIN_VALUE;
    for (int num : arr) {
      if (num > first) {
        second = first;
        first = num;
      } else if (num > second && num != first) {
        second = num;
      }
    }
    System.out.println(second);
  }
```

```
}
```

18. Swap Two Numbers

```
public class SwapNumbers {
  public static void main(String[] args) {
    int a = 5, b = 10;
    a = a + b;
    b = a - b;
    a = a - b;
    System.out.println("a: " + a + ", b: " + b);
  }
}
```

19. Print the Pascal's Triangle

```
public class PascalsTriangle {
  public static void main(String[] args) {
    int rows = 5;
    for (int i = 0; i < rows; i++) {
      int num = 1;
      System.out.format("%" + (rows - i) * 2 + "s", "");
      for (int j = 0; j <= i; j++) {
        System.out.format("%4d", num);
        num = num * (i - j) / (j + 1);
      }
      System.out.println();
    }
  }
}
```

20. Find the Missing Number in an Array

```
public class MissingNumber {
  public static void main(String[] args) {
```

```
    int[] arr = {1, 2, 4, 5, 6};
    int n = arr.length + 1;
    int total = n * (n + 1) / 2;
    for (int num : arr) {
      total -= num;
    }
    System.out.println(total);
  }
}
```

21. Convert Decimal to Binary

```
public class DecimalToBinary {
  public static void main(String[] args) {
    int num = 10;
    String binary = Integer.toBinaryString(num);
    System.out.println(binary);
  }
}
```

22. Check for Perfect Number

```
public class PerfectNumber {
  public static void main(String[] args) {
    int num = 28, sum = 0;
    for (int i = 1; i <= num / 2; i++) {
      if (num % i == 0) {
        sum += i;
      }
    }
    System.out.println(num == sum);
  }
}
```

23. Implementing a Simple Calculator

```
import java.util.Scanner;

public class SimpleCalculator {

  public static void main(String[] args) {

    Scanner scanner = new Scanner(System.in);

    System.out.print("Enter first number: ");

    double num1 = scanner.nextDouble();

    System.out.print("Enter second number: ");

    double num2 = scanner.nextDouble();

    System.out.print("Enter operation (+, -, *, /): ");

    char operation = scanner.next().charAt(0);

    double result;

    switch (operation) {

      case '+': result = num1 + num2; break;

      case '-': result = num1 - num2; break;

      case '*': result = num1 * num2; break;

      case '/': result = num1 / num2; break;

      default: throw new IllegalArgumentException("Invalid operation");

    }

    System.out.println("Result: " + result);

  }

}
```

24. Find the Sum of Digits of a Number

```
public class SumOfDigits {

  public static void main(String[] args) {

    int num = 12345, sum = 0;

    while (num != 0) {

      sum += num % 10;

      num /= 10;

    }
```

```
    System.out.println(sum);
  }
}
```

25. Find the Length of a String

```
public class StringLength {
  public static void main(String[] args) {
    String str = "Automation";
    System.out.println(str.length());
  }
}
```

26. Check if a String is Empty

```
public class CheckEmptyString {
  public static void main(String[] args) {
    String str = "";
    System.out.println(str.isEmpty());
  }
}
```

27. Count the Occurrences of a Character in a String

```
public class CountCharacter {
  public static void main(String[] args) {
    String str = "Automation";
    char ch = 'a';
    int count = 0;
    for (char c : str.toCharArray()) {
      if (c == ch) count++;
    }
    System.out.println(count);
  }
}
```

28. Find the First Non-Repeated Character in a String

```
import java.util.LinkedHashMap;

import java.util.Map;

public class FirstNonRepeatedCharacter {

  public static void main(String[] args) {

    String str = "swiss";

    Map<Character, Integer> charCount = new LinkedHashMap<>();

    for (char c : str.toCharArray()) {

      charCount.put(c, charCount.getOrDefault(c, 0) + 1);

    }

    for (Map.Entry<Character, Integer> entry : charCount.entrySet()) {

      if (entry.getValue() == 1) {

        System.out.println(entry.getKey());

        break;

      }

    }

  }

}
```

29. Remove All Whitespaces from a String

```
public class RemoveWhitespaces {

  public static void main(String[] args) {

    String str = " A u t o m a t i o n ";

    String result = str.replaceAll("\\s+", "");

    System.out.println(result);

  }

}
```

30. Find the Common Elements in Two Arrays

```
import java.util.HashSet;

public class CommonElements {
```

```
  public static void main(String[] args) {
    int[] arr1 = {1, 2, 3, 4};
    int[] arr2 = {3, 4, 5, 6};
    HashSet<Integer> set = new HashSet<>();
    for (int num : arr1) {
      set.add(num);
    }
    for (int num : arr2) {
      if (set.contains(num)) {
        System.out.print(num + " ");
      }
    }
  }
}
```

31. Find the Factorial of a Number using Recursion

```
public class FactorialRecursion {
  public static void main(String[] args) {
    int num = 5;
    System.out.println(factorial(num));
  }

  static int factorial(int n) {
    if (n == 0) return 1;
    return n * factorial(n - 1);
  }
}
```

32. Generate Random Numbers

```
import java.util.Random;
public class RandomNumbers {
```

```
  public static void main(String[] args) {
    Random random = new Random();
    for (int i = 0; i < 5; i++) {
      System.out.println(random.nextInt(100)); // Random number between 0-99
    }
  }
}
```

33. Check if a Year is Leap Year

```
public class LeapYear {
  public static void main(String[] args) {
    int year = 2024;
    boolean isLeap = (year % 4 == 0 && year % 100 != 0) || (year % 400 == 0);
    System.out.println(isLeap);
  }
}
```

34. Find the Sum of First N Natural Numbers

```
public class SumOfNaturalNumbers {
  public static void main(String[] args) {
    int n = 10, sum = n * (n + 1) / 2;
    System.out.println(sum);
  }
}
```

35. Implement a Simple Login System

```
import java.util.Scanner;
public class SimpleLogin {
  public static void main(String[] args) {
    String username = "admin";
    String password = "password";
    Scanner scanner = new Scanner(System.in);
```

```
    System.out.print("Enter username: ");
    String inputUsername = scanner.nextLine();
    System.out.print("Enter password: ");
    String inputPassword = scanner.nextLine();
    if (username.equals(inputUsername) && password.equals(inputPassword)) {
      System.out.println("Login successful!");
    } else {
      System.out.println("Login failed!");
    }
  }
}
```

36. Check if a String Contains Another String

```
public class StringContains {
  public static void main(String[] args) {
    String str1 = "Automation Testing";
    String str2 = "Testing";
    System.out.println(str1.contains(str2));
  }
}
```

37. Find the Maximum Occurring Character in a String

```
import java.util.HashMap;
public class MaxOccurringCharacter {
  public static void main(String[] args) {
    String str = "programming";
    HashMap<Character, Integer> charCount = new HashMap<>();
    for (char c : str.toCharArray()) {
      charCount.put(c, charCount.getOrDefault(c, 0) + 1);
    }
    char maxChar = str.charAt(0);
```

```
    int maxCount = 0;
    for (char c : charCount.keySet()) {
      if (charCount.get(c) > maxCount) {
        maxCount = charCount.get(c);
        maxChar = c;
      }
    }
    System.out.println(maxChar);
  }
}
```

38. Implementing Bubble Sort

```
public class BubbleSort {
  public static void main(String[] args) {
    int[] arr = {64, 34, 25, 12, 22, 11, 90};
    int n = arr.length;
    for (int i = 0; i < n - 1; i++) {
      for (int j = 0; j < n - i - 1; j++) {
        if (arr[j] > arr[j + 1]) {
          int temp = arr[j];
          arr[j] = arr[j + 1];
          arr[j + 1] = temp;
        }
      }
    }
    for (int num : arr) {
      System.out.print(num + " ");
    }
  }
}
```

39. Implementing Selection Sort

```
public class SelectionSort {
  public static void main(String[] args) {
    int[] arr = {64, 25, 12, 22, 11};
    int n = arr.length;
    for (int i = 0; i < n - 1; i++) {
      int minIndex = i;
      for (int j = i + 1; j < n; j++) {
        if (arr[j] < arr[minIndex]) {
          minIndex = j;
        }
      }
      int temp = arr[minIndex];
      arr[minIndex] = arr[i];
      arr[i] = temp;
    }
    for (int num : arr) {
      System.out.print(num + " ");
    }
  }
}
```